

The Impact of Cigarette Smoking on Hypertension: A 24-Year Case-Control Study from the Framingham Heart Study

Group 17 - Author contributions are the same

T. Ghaneh, M. Loni, X. Zheng

Abstract

This case-control study investigates the relationship between cigarette smoking (as a binary exposure) and the risk of hypertension (as a binary outcome) over 24 years, using data from the Framingham Heart Study. The Framingham Heart Study is a long-term cohort study on cardiovascular diseases conducted in the community of Framingham, Massachusetts. This analysis uses data from the first examination period and the corresponding results after 24 years. The study aims to estimate the causal effect of smoking on the risk of hypertension. Hypertension is defined as receiving treatment for high blood pressure or having a systolic blood pressure ≥ 140 mmHg or a diastolic blood pressure ≥ 90 mmHg. In addition, we assumed that smoking would reduce the risk of hypertension, but this assumption is not fully confirmed in this study.

1 Introduction

This study aims to assess the causal effect of cigarette smoking on the risk of developing hypertension over a 24-year period using data from the Framingham Heart Study. Hypertension is a major contributor to cardiovascular diseases. Similarly, cigarette smoking is a powerful cardiovascular risk factor that elevates the risk of heart disease, stroke, and other related conditions. Despite well-documented evidence of smoking's adverse effects on cardiovascular health, the causal relationship between smoking and hypertension remains unclear and chronic smoking's effect on blood pressure regulation is less understood. To address this gap, we formulate the hypothesis that: "Current cigarette smoking causally increases the long-term risk of hypertension."

2 Methods

2.1 Study Design

The data source for this study is the Framingham Heart Study, a long-term cohort study on cardiovascular diseases among individuals in the community of Framingham, Massachusetts, which began in 1948. This study utilizes a subset of data from that study. The dataset includes laboratory and questionnaire data measured at baseline for 4,434 participants. Additionally, it contains clinical and event data collected over a 24-year follow-up period. Clinical data were collected during three examinations, each conducted approximately six years apart. However, only the data from the 24-year follow-up period were analyzed in this study, while intermediate examination results were not considered.

This study investigates the causal relationship between current cigarette smoking (0 = Not current smoker, 1 = Current smoker) and hypertension, which is defined as either receiving treatment for high blood pressure or having a systolic blood pressure of at least 140 mmHg or a diastolic blood pressure of at least 90 mmHg. To evaluate this relationship, the average treatment effect (ATE) was estimated to assess the overall impact of current cigarette smoking on hypertension (X=0: Smoker, X=1: non-Smoker; Y(1) represents potential hypertension status if smoking (1=hypertensive, 0=no hypertensive), Y(0) represents potential hypertension status if not smoking). Additionally, the average treatment effect on the treated (ATT) was estimated to measure the effect of smoking on hypertension incidence specifically among current smokers.

$$\text{ATE} = E[Y(1) - Y(0)] \quad (1)$$

$$\text{ATT} = E[Y(1) - Y(0) \mid X = 1] \quad (2)$$

2.2 Directed Acyclic Graph

Since this study is based on observational data rather than data from a randomized controlled trial (RCT), causal relationships cannot be directly inferred. This limitation arises because key assumptions necessary for causal inference—namely, consistency, exchangeability, and positivity—may not be fully satisfied. Specifically, the consistency assumption requires a precise and consistent definition of current cigarette smoking. The exchangeability assumption necessitates that, after adjusting for confounders, the distributions of smokers and non-smokers remain comparable. Furthermore, the positivity assumption ensures that all individuals have a nonzero probability of being either a smoker or a non-smoker. By adjusting for confounders that influence both smoking and hypertension, a more reliable estimate of the causal effect can be obtained.

To identify confounders affecting the relationship between current cigarette smoking and hypertension, a Directed Acyclic Graph (DAG) was constructed. The DAG provides a structured visual representation of the causal relationships and helps determine which confounders require adjustment. In particular, the backdoor criterion was applied to identify the adjustment set, thereby improving estimation efficiency. Step-wise model selection and cross-validation were also used to verify the model selection.

2.3 Propensity Score Method

Once the confounders were identified, a propensity score model was used to characterize the relationship between confounders and current cigarette smoking. Subsequently, inverse probability weighting (IPW) was applied to correct for data imbalance. To ensure the validity of using the propensity score method, the distributions of propensity scores were examined in both the smoker and non-smoker groups, in order to verify that the exchangeability and positivity assumptions were met. After reweighting, standardized mean differences (SMD) and Love plots were utilized to assess whether the data were sufficiently balanced and whether the exchangeability assumption was satisfied (SMD < 0.1).

Once balance was achieved, a regression model, such as a weighted logistic regression, was employed to estimate the causal effect of smoking on hypertension. The model incorporated current cigarette smoking and the identified confounders as covariates. Additionally, mediating variables were considered to further explore potential mechanisms underlying the relationship between smoking and hypertension. By including these mediators, the analysis aimed to provide a more comprehensive understanding of how smoking contributes to hypertension development. Finally, ATE and ATT were computed to quantify the unbiased causal effect of current cigarette smoking on hypertension.

The propensity score represents the probability that an individual will receive treatment given a set of covariates. By applying matching, stratification, or weighting based on this probability, the treatment and control groups can be made comparable with respect to confounders. In this study, the propensity score method was chosen over G-computation and other direct modeling approaches due to the unbalanced distribution of the outcome variable, hypertension. This imbalance could potentially bias causal effect estimates. However, by using propensity score weighting, the impact of this distributional imbalance was minimized, thereby improving the robustness and reliability of the estimation.

3 Results

3.1 Data exploration

As shown in Table 1, 31% of smokers have hypertension, while 42% of non-smokers have hypertension.

Table 1: Proportions of Hypertension by Smoking Status

Current Smoking	Proportion by Hypertension Status	
	0 (No Hypertension)	1 (Hypertension)
0 (Non-smoker)	0.5801	0.4199
1 (Current smoker)	0.6914	0.3086

Among 4,241 observations, 194 had missing values, primarily in Education, BPMEDS (use of antihypertensive medication), and BMI. Little’s MCAR test ($p = 1.14 \times 10^{-6}$) rejected the MCAR assumption. Since the missingness was likely MAR, we used likelihood based method instead of complete-case analysis to avoid bias and efficiency loss.

3.2 Directed Acyclic Graph

By analyzing the DAG graph, three confounders were identified: sex, age, diabetes and education.

3.3 Propensity Score Model and weight

First, the causal hypothesis needs to be examined for the purposes of identifiability. To test the validity of the Positivity hypothesis, some density plots are plotted. Sufficient

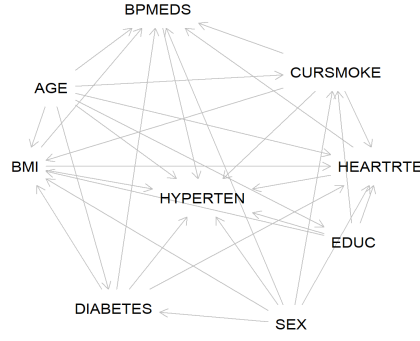


Figure 1: Directed Acyclic Graph (DAG)

acceptable overlap and non-zero probability of the variable age indicate that the positivity is true.

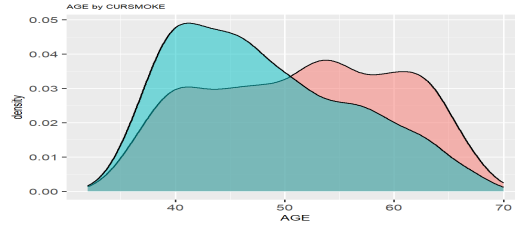


Figure 2: Density plot of AGE
(blue: non-smokers, red: smokers)

Then to check for possible exchangeability, refer to the scale bar chart below. These graphs attempt to answer the question of whether smoking group and non-smoking group are interchangeable within each confounder. The figure shows that this exchangeability is acceptable at the level of education, but not at the other covariates.

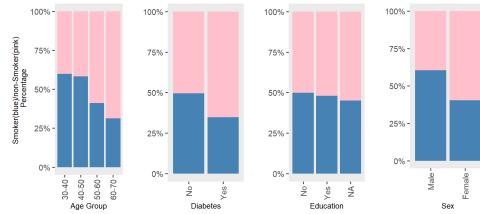


Figure 3: Bar plots to check exchangeability

After applying IPW to the data using the propensity model, the smoking and non-smoking groups reached balanced on the covariates (Figure 4).

Starting with an elaborate logistic regression model that included higher-order terms and interactions, the main challenge was finding a balance between two competing priorities: ensuring the significance, interpretability, and simplicity of parameter estimates while also enhancing the model's robustness by accounting for as many potential confounders as possible. Eventually, the best decision made by integrating all above discussions is a logistic regression that models HYPERTEN risk against CURS MOKE exposure and potential confounders AGE, DIABETES, EDUC, SEX, and the interaction SEX:AGE.

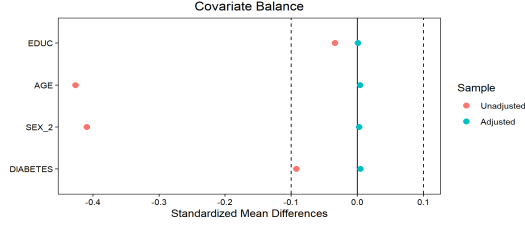


Figure 4: Love plot

Table 2: Logistic Regression Results

Variable	Estimate	Std. Error	p-value
Intercept	1.457	0.658	0.027*
Current Smoker (CURSMOKE)	-0.222	0.070	0.002**
Age (AGE)	-0.036	0.013	0.005**
Diabetes (DIABETES)	0.535	0.199	0.007**
Education (EDUC)	-0.241	0.076	0.002**
Sex (SEX)	-3.642	0.420	<0.001***
Age:Sex Interaction (AGE:SEX)	0.071	0.008	<0.001***
AIC		5,165.3	

*** p<0.001, ** p<0.01, * p<0.05.

Model specification: $HYPERTEN \sim CURSMOKE + AGE + DIABETES + EDUC + SEX + SEX \times AGE$
Null deviance: 5,671.1 (df=4,320), Residual deviance: 5,151.3 (df=4,314)

When comparing smokers to non-smokers, the estimated risk difference (RD) of -4.5% (95% CI: -7.4%, -1.7%) shows a statistically significant decrease in the prevalence of hypertension. The exclusion of zero from the confidence interval confirms this protective association is unlikely to result from random variation.

The ATT estimate of -0.045 (95% CI: -7.3%, -1.7%) indicates a statistically significant protective association between smoking and the risk of hypertension among smokers.

4 Discussion and Conclusion

The results contradicted our assumption by suggesting a negative correlation between smoking and the risk of hypertension. There might be significant confounding variables: measurements of the BPMEDS variable in the available data may not completely reflect the possibility of systemic variations in drug use among smokers, where smokers use more antihypertensive drugs as a result of more frequent doctor visits.

Second, the study's use of smoking/non-smoking dichotomous exposure variables may obscure important heterogeneity, specifically by failing to identify subgroup variations in smoking intensity (e.g., daily smoking) and concurrent drug use. It is recommended that future research use a three-categorical variable system and a more precise description of exposure in order to address this issue: 1) nonsmokers; 2) smokers without antihypertensive medication; and 3) smokers and antihypertensive medication users. Meanwhile, by gathering comprehensive medication records and using the instrumental variable approach, the possible residual confounding should be substantially reduced. This enhancement will make it easier to understand the actual causal effect between smoking and the risk of hypertension.